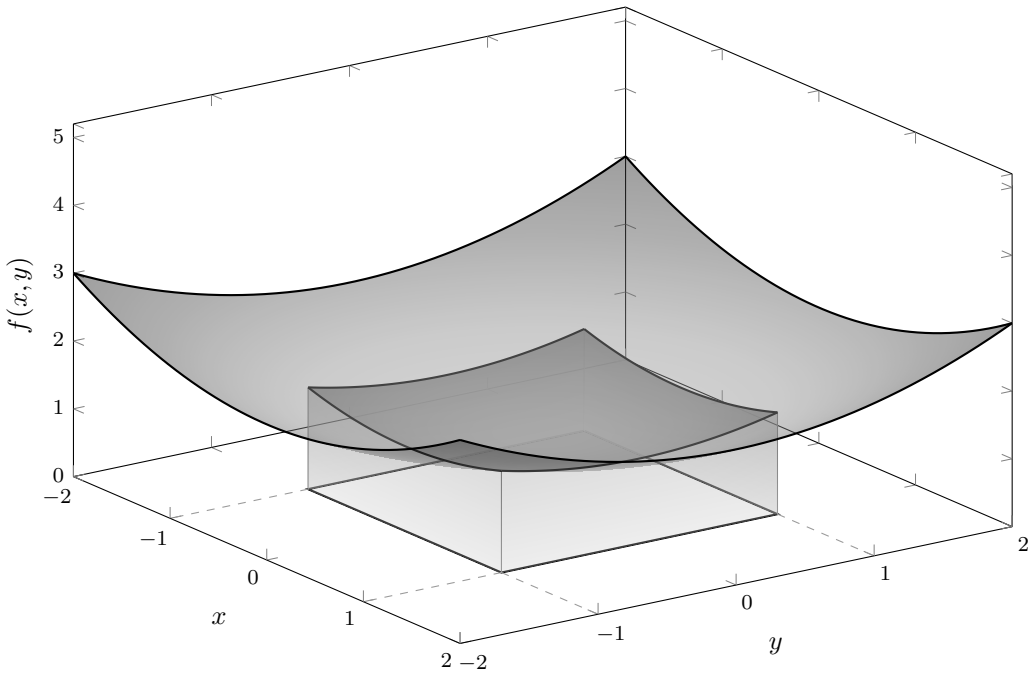
Multiple Integrals

Multiple integrals are a way of generalising integral under the curve to higher dimensions. A double integral is an integral of a function with two variables $f(x, y)$ and is now a volume under a surface rather than an area under a line Consider the diagram below for the function $f(x, y) = \frac{1}{4}(x^2 + y^2) + 1$



The diagram shows the area under the function $f(x, y)$ on the domain $[-1, 1] \times [-1, 1]$. The expression for this integral can be written in the following ways

$$V = \iint_{[-1,1]\times[-1,1]} \left(\frac{1}{4}(x^2 + y^2) + 1\right) dx\, dy$$
$$V = \int_{-1}^{1}\int_{-1}^{1} \left(\frac{1}{4}(x^2 + y^2) + 1\right) dx\, dy$$

Notice there are now two integral symbols, which are there to signify that we integrate with respect to two variables. Alongside the second integral, a dy must be included in addition to the dx. Evaluating double integrals has no major conceptual leaps if you understand integration with one variable and the concept of fixing one variable − the same as from partial derivatives. In order to calculate the definite integral, first calculate the inside integral while treating y as constant. It should be noted that the bounds are to be substituted into the x variable, since the dx is on the inside integral. Secondly, evaluate the outside

integral as you would a regular integral. A demonstration of this process follows

$$\begin{aligned} V &= \int_{-1}^{1}\int_{-1}^{1}\left(\frac{1}{4}(x^2+y^2)+1\right)dx\,dy \\ &= \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}(x^2+y^2+4)\,dx\,dy \\ &= \frac{1}{4}\int_{-1}^{1}\left[\frac{1}{3}x^3+xy^2+4x\right]_{x=-1}^{x=1}dy \end{aligned}$$

Taking a quick aside to evaluate the integrand given the bounds of x

$$\begin{aligned} \left[\frac{1}{3}x^3+xy^2+4x\right]_{x=-1}^{x=1} &= \left(\frac{1}{3}(1)^3+1\cdot y^2+4\cdot 1\right) \\ &\quad -\left(\frac{1}{3}(-1)^3+(-1)\cdot y^2+4\cdot(-1)\right) \\ &= 2y^2+\frac{26}{3} \end{aligned}$$

Returning to the double integral

$$\begin{aligned} V &= \frac{1}{4}\int_{-1}^{1}\left(2y^2+\frac{26}{3}\right)dy \\ &= \frac{1}{4}\left[\frac{2}{3}y^3+\frac{26}{3}y\right]_{y=-1}^{y=1} \\ &= \frac{1}{4}\left(\frac{2}{3}\cdot 1^3+\frac{26}{3}\cdot 1\right)-\frac{1}{4}\left(\frac{2}{3}\cdot(-1)^3+\frac{26}{3}\cdot(-1)\right) \\ &= \frac{14}{3} \end{aligned}$$

Integrating Over Non-Rectangular Regions

Before integrating non-rectangular domains, we will quickly discuss the concept of the bounds of an integral.

Consider the following integral

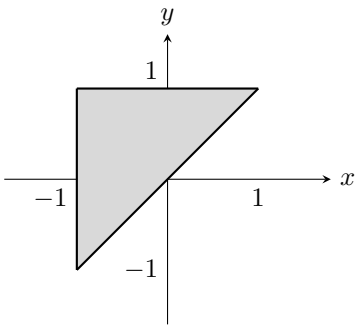
$$\int_{-1}^{1} x^2 \, dx$$

This expression when spoken verbally will typically be read *the integral of* x^2 *from -1 to 1.* It is important to make the conceptual clarification that the integral is being performed *on the region [-1,1].* Likewise, the following double integral

$$V = \int_{x=-2}^{x=0} \int_{y=-1}^{y=1} \frac{1}{4}(x^2 - y^2) \, dy \, dx$$

is being performed over the region $\{-2 \leq x \leq 0, \ -1 \leq y \leq 1\}$. This concept is particularly important when dealing with double integrals because it makes the concept of integrating over non-rectangular domains easier.

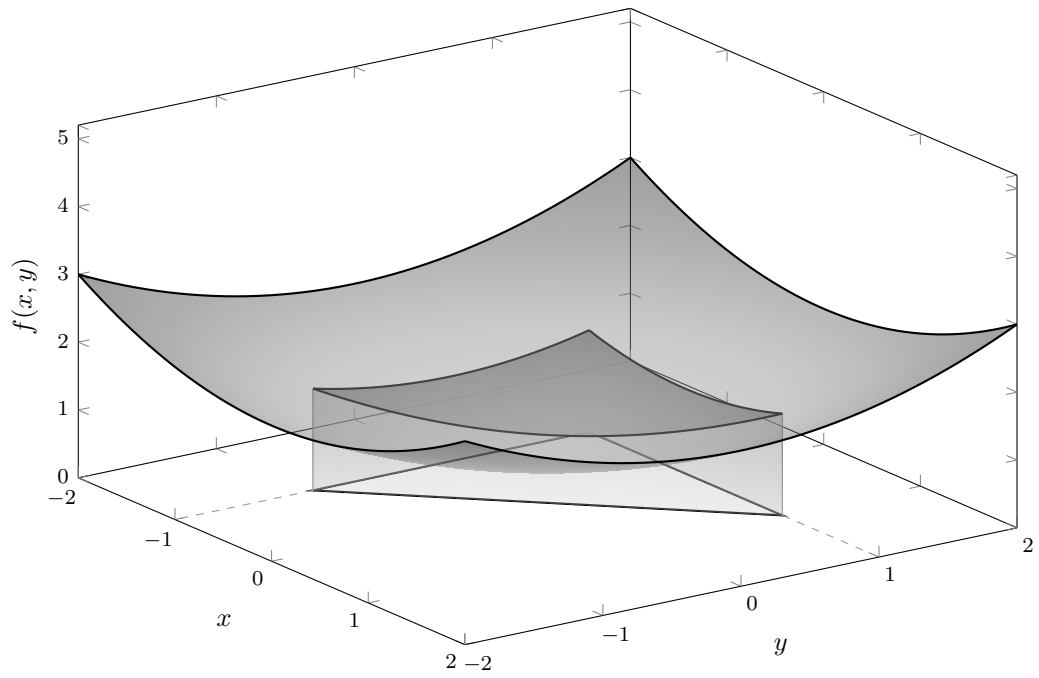
Consider the following region in the $x - y$ plane



Unlike the rectangular region $\{-2 \leq x \leq 0, \ -1 \leq y \leq 1\}$, the upper and lower $y-$bounds cannot both be fixed numbers. The lower $y-$bound changes with x, in particular the highlighted region sits above the line $y = x$, so the expression for the region must rely on using the x variable for lower bound for the y variable, giving $x \leq y$. Given these constraints, the expression for the region is $\{-1 \leq x \leq 1, \ x \leq y \leq 1\}$. So, if we copy the upper and lower bounds given in the expression and use them in the bounds for a double integral below

$$\int_{x=-1}^{x=1} \int_{y=x}^{y=1} \Big(\frac{1}{4}(x^2 + y^2) + 1 \Big) dy \, dx$$

we can integrate our function $f(x, y)$ over a region of this shape



and it evaluates the exact same as the example double integral at the beginning. A keen individual could try calculate this for them self and verify that this equals $\frac{14}{6}$ (half the value of the worked example)